TransTech Systems, Inc.



PQI 380go

Pavement Quality Indicator

Quick Start Guide

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Case



PQI 380go



DC Cord



AC Cord



Handle



Charger

Battery Charging & Care



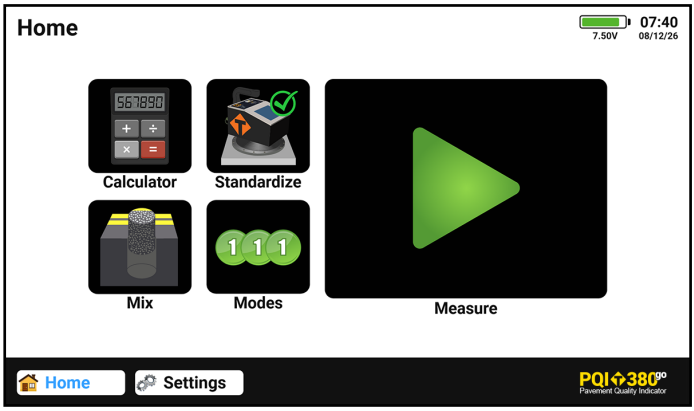
1. Turn the PQI 380go unit OFF.
2. Connect the charger to the charger connector located on the back of the PQI 380go.
3. Plug the charger into a standard AC outlet, verify red LED indicator is on.
4. The red indicator lamp will turn green to indicate that the batteries are charged. (approx. 4hrs)
5. Unplug the charger from the power source before disconnecting the charger from the PQI 380go.

Battery Care

- Whenever uncertain about the battery charge level or condition, recharge it
- The battery will self-discharge and should be recharged at least every 30 days
- An occasional complete discharge followed by a full recharge is recommended
- **NEVER** drop the battery as this can damage the internals
- **DO NOT** store in freezer or expose to extreme heat
- Battery should be fully charged before use

Starting the Software

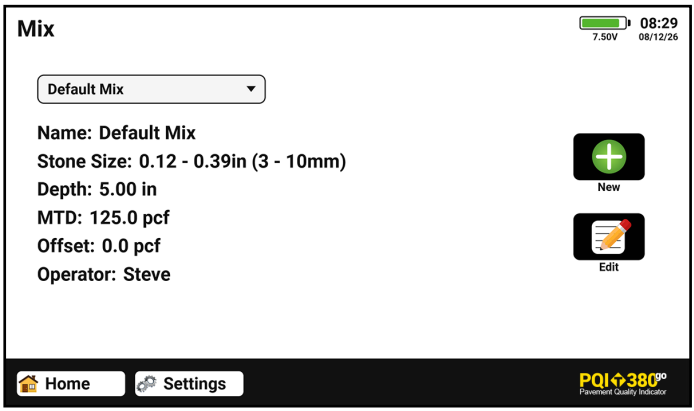
After powering on, the PQI 380go initializes for 30 seconds while displaying multiple splash screens. At the end of boot up the home screen is displayed.



Mix Details

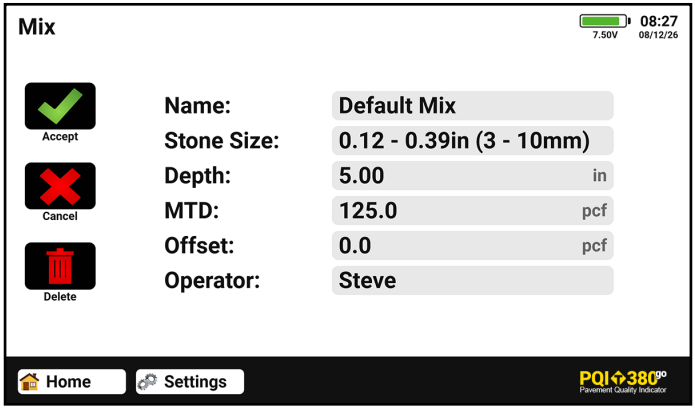
From the Home screen, tap **Mix**. The mix listed in the drop down menu on the left is displayed in detail directly below. To edit the details of this mix, tap **Edit**.

Note: Readings taken prior to accurately setting up the mix details will result in incorrect density and compaction results.

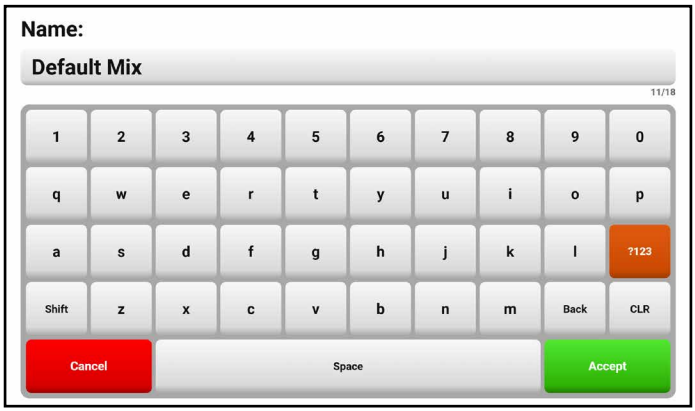


Edit Mix - Name

In the Edit Mix screen there are six fields labeled **Name, Stone Size, Depth, MTD, Offset** and **Operator**. By tapping one of these fields, you will enter the corresponding screen which will allow you to edit and save information for that mix value. For example tap the gray shaded **Name** field.

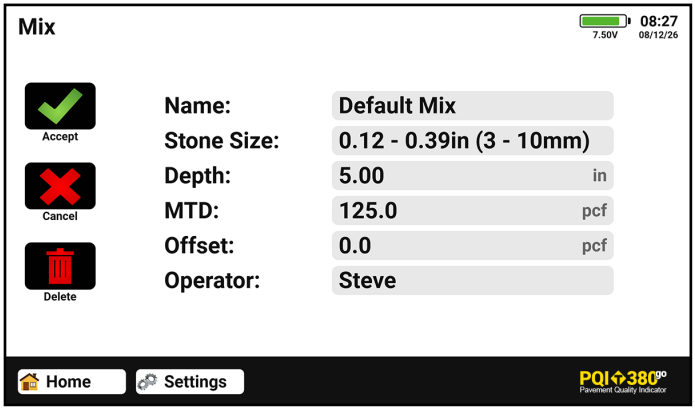


The keyboard will pop up allowing you to change the mix name. Tap **"CLR"** if you would like to start over with a new name. Tap **Shift** to toggle from lowercase to uppercase letters. Once editing has been completed, tap **Accept** to return to the Edit Mix screen.

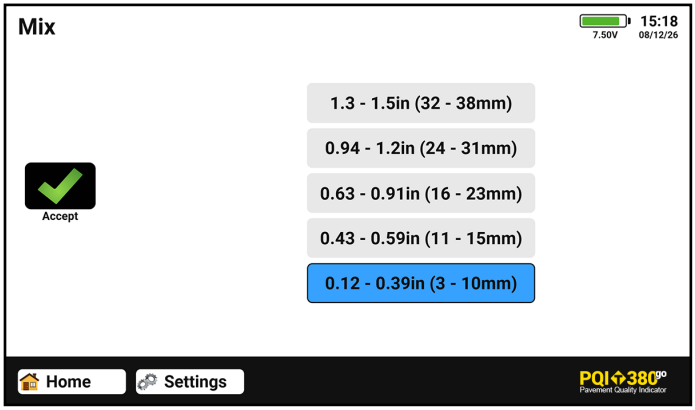


Edit Mix - Stone Size

Tap the **Stone Size** field. There are five options for aggregate sizes listed in inches (millimeters). If, however, you do not see your specific stone size listed choose the closest stone size available.

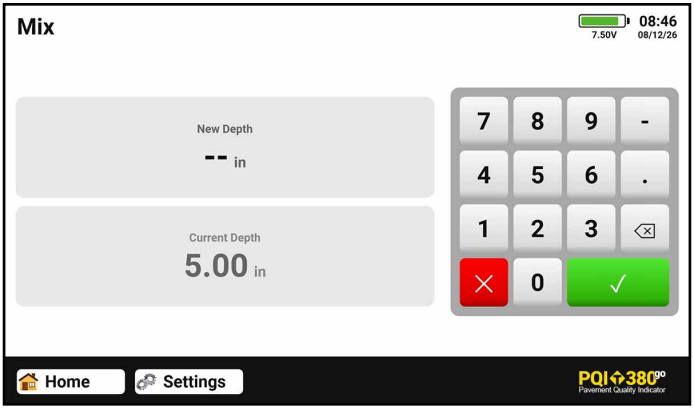


For example, if your mix has an aggregate size of 0.79in (20mm), select **0.63 - 0.91in (16mm - 23mm)**. Tap **Accept** after you made your selection to return to the Edit Mix screen.

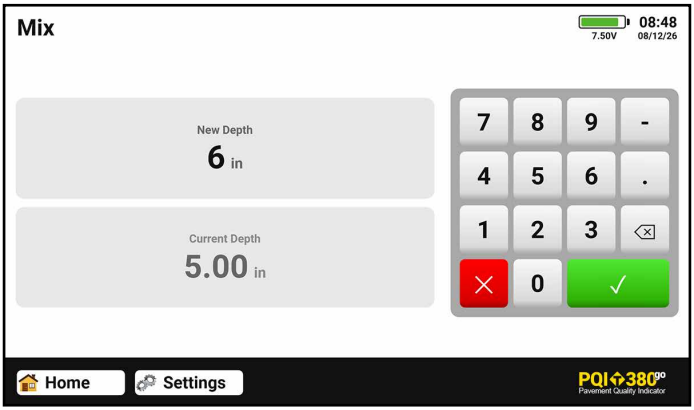


Edit Mix - Depth

Tap the **Depth** field from the Edit Mix screen to enter the Depth screen, then tap the **New Depth** field.



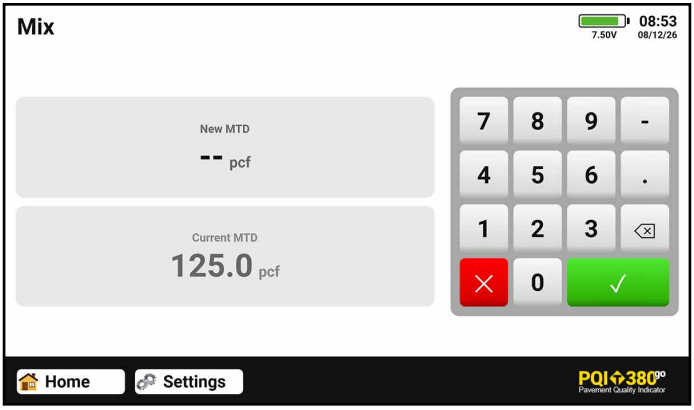
Enter the depth of the asphalt mat on the numeric keypad in the units you previously selected for the depth. Once the correct value has been entered, tap **Accept** to return to the Depth screen.



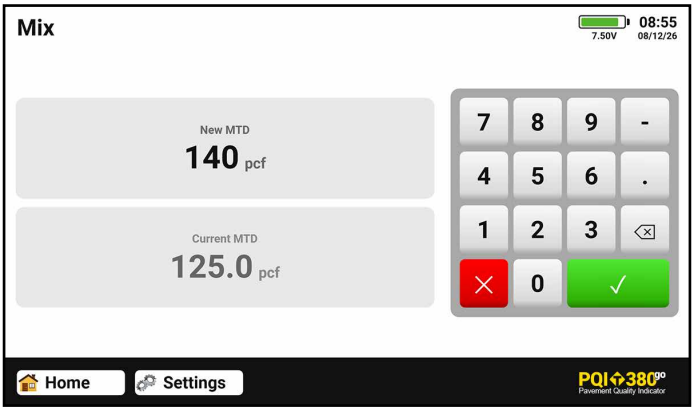
An out of range Warning will display for depths entered outside of the following range: 0.75in - 6.0in (19.05mm - 152.4mm)

Edit Mix - MTD (Max Theoretical Density)

Tap the **MTD** field from the Edit Mix screen to enter the MTD screen, then tap the **New MTD** field.



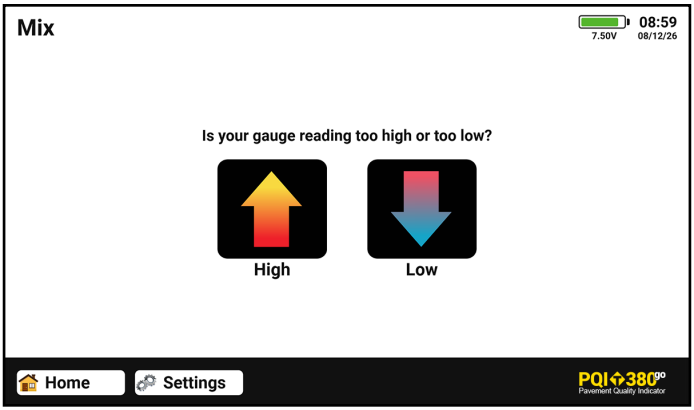
Enter the **Maximum Theoretical Density (MTD)** of the mix on the numeric keypad in the units you previously selected. The MTD is provided from the asphalt mix designer and is a key value in determining the percent compaction. Once the correct value has been entered, tap **Accept** to return to the MTD screen.



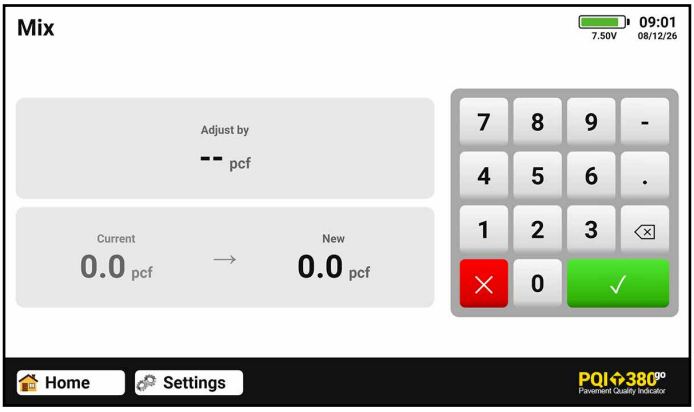
An out of range Warning will display for an MTD entered outside of the following range: 100lb/ft³ - 200lb/ft³ (1601.8kg/m³ - 3203.7kg/m³)

Edit Mix - Offset

The offset for the mix can be adjusted at a later time. **Measurements taken prior to adjusting the offset will not include the offset.**

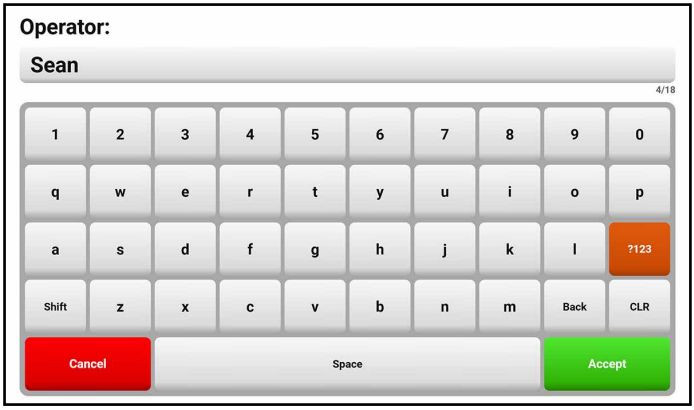


You will need to take a measurement to determine the offset. The offset feature will be revisited later in the **calculating the offset** section of this quick start guide.

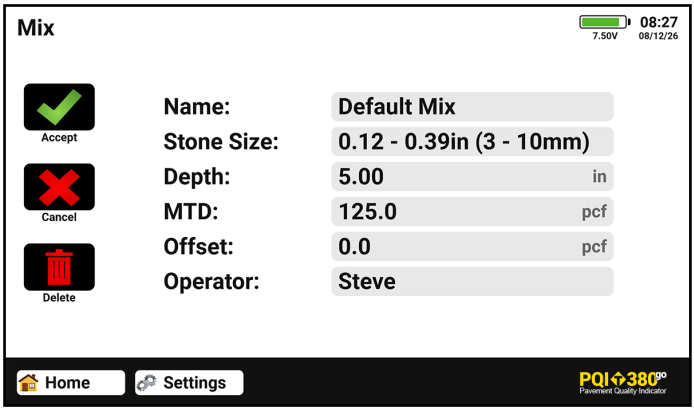


Edit Mix - Operator

Tap the **Operator** field from the Edit Mix screen. The keyboard will pop up allowing you to change the operator's name. Enter your name as the operator of the gauge, then tap **Accept** to return to the Edit Mix screen.

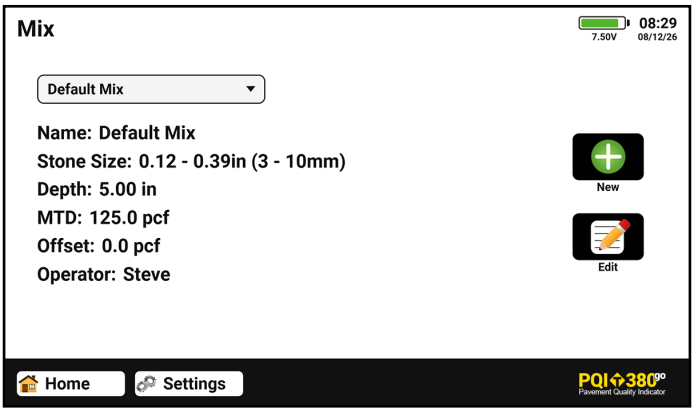


Verify all of the mix information on this screen. Once you have verified the results of the mix and are satisfied with the values, tap **Accept** to save the mix.

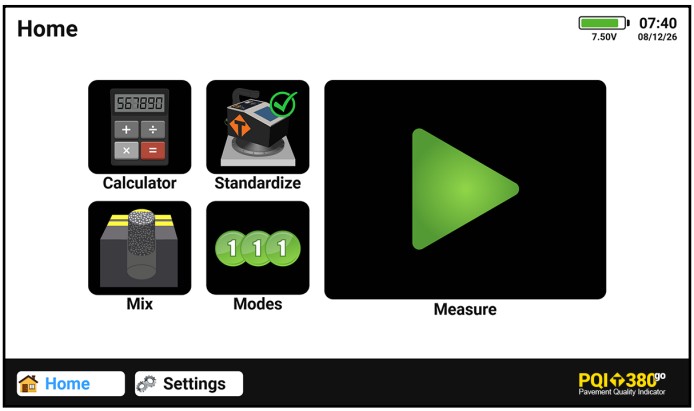


Edit Mix

After exiting the Edit Mix screen, the PQI 380go will return to the Mix screen. Here you can tap the **New** button to create a new mix, tap the **Edit** button to edit the currently selected mix or use the drop down menu to select a previously created mix.

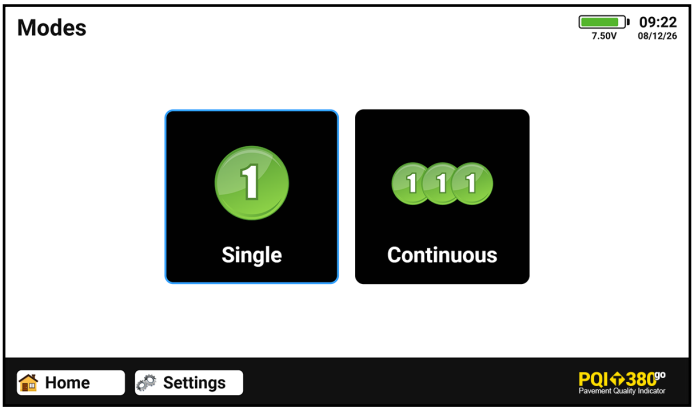


The mix shown is the **CURRENT MIX** that the gauge will use when taking readings. Prior to exiting this screen, be sure this is the mix you want and the information been verified. Tap **Home** on the task bar to return to the Home screen. If the gauge is shutdown, the current mix prior to shutdown will remain the current mix when started back up.



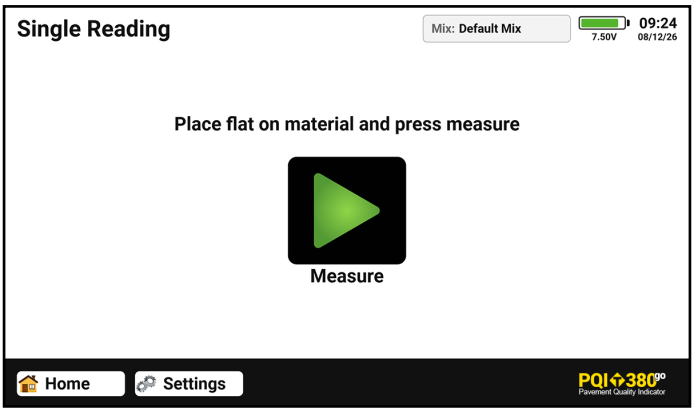
Reading Modes

The PQI 380+ has two reading modes - **Single and Continuous**. **Data will NOT be saved on the PQI 380go, it is up to the operator to document the results.** To select or change a reading mode tap **Modes** from the Home screen. Then tap the mode you would like to use. Once selected the active mode will be highlighted with a blue border and automatically redirect you to the measurement screen awaiting a reading. Continuous mode will start immediately.



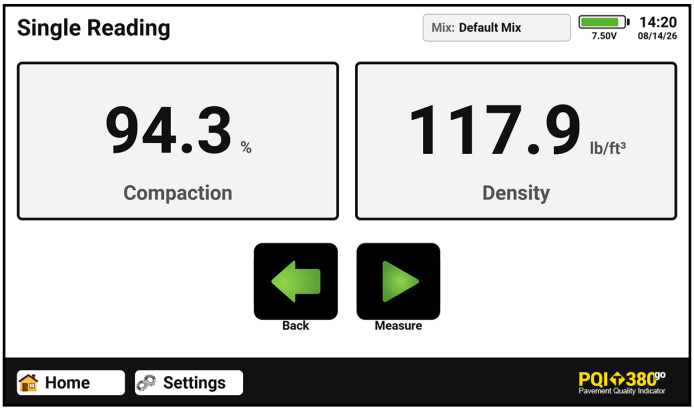
Reading Modes - Single

From the Home screen, tap **Modes**, then tap **Single**.



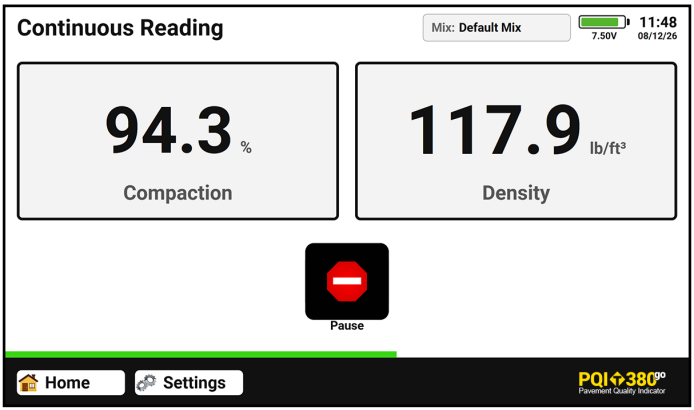
Reading Modes - Single

Press **Measure** on the Single Reading screen. The PQI 380go will display the results. **Data will NOT be saved on the PQI 380go, it is up to the operator to document the results.** Press **Measure** to take another reading or press **Back** to return to the Single Reading start screen.



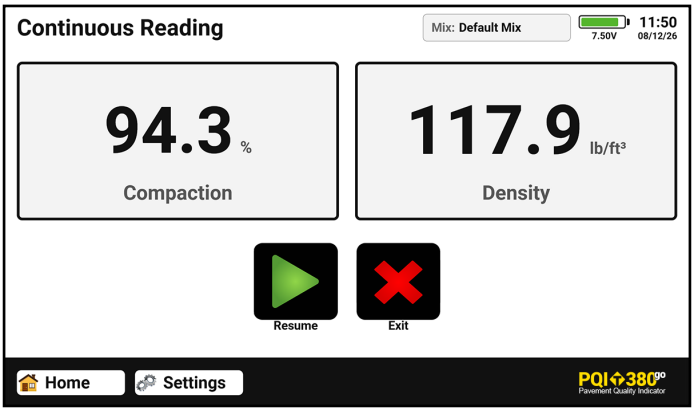
Reading Modes - Continuous

From the Home screen, tap **Modes**, then tap **Continuous**. The PQI 380go will immediately start taking measurements every three seconds.



Reading Modes - Continuous

Data will NOT be saved on the PQI 380go, it is up to the operator to document the results. Press **Pause** if needed to write down the results. Press **Resume** when you are ready to continue taking measurements or press **Exit** to return to the Home screen.



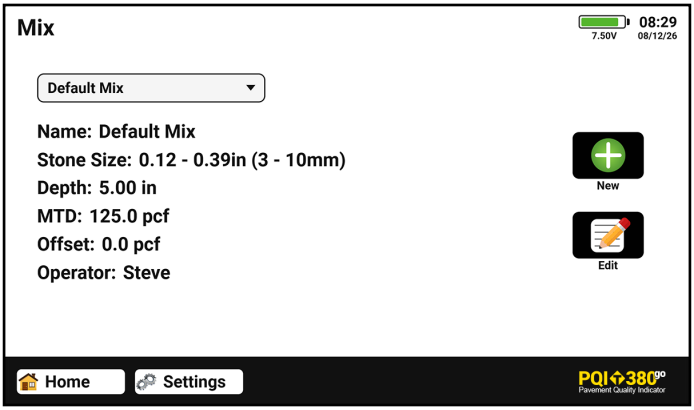
Calculating the Offset

Direct percent compaction of the test mat is achieved by obtaining physical samples (cores) from the locations previously tested. Calculate the difference between the average PQI 380go readings and the core density values. This difference will be the Offset for that specific mix that will be stored in the gauge for that mix. Therefore, you will be adjusting the value of the PQI 380go readings by that amount in order for the gauge to read the same as the core(s).

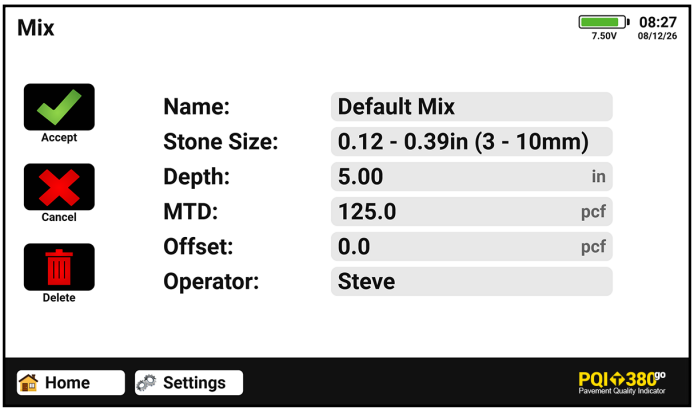
Once the numeric difference is calculated, determine whether the PQI 380go is reading **too high** or **too low**. For example, if the PQI 380go reads **155lb/ft³**, and you would like it to read **150lb/ft³**, the PQI 380go is reading too high and **the adjusted value for the offset is 5**.

PQI 380go Offset Setup

From the Home screen, tap **Mix**. **Be sure that the selected mix is the mix you wish to add the offset to.**

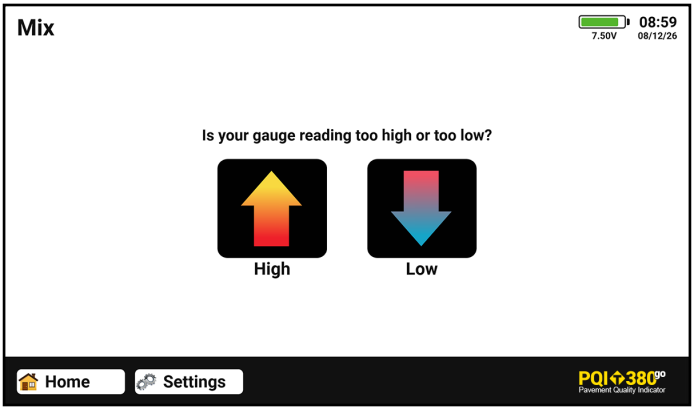


Tap **Edit**, then tap the gray shaded **Offset** field.

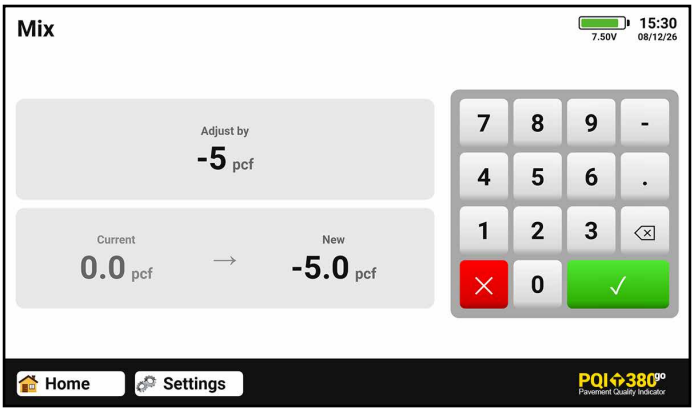


PQI 380+ Offset Setup

In the previous step we determined that the PQI 380go is reading too high and the adjusted value for the offset is 5. Tap **HIGH** to continue.

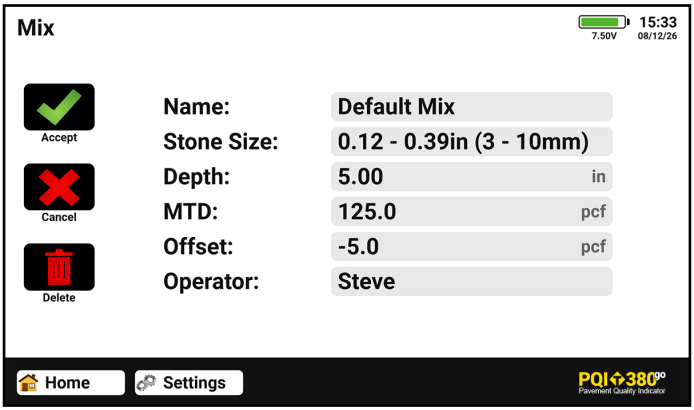


The offset adjustment will display the current offset value. Enter the adjusted value for the offset which we previously determined would be 5, then tap the **green checkmark** to return to the **Edit Mix** screen.

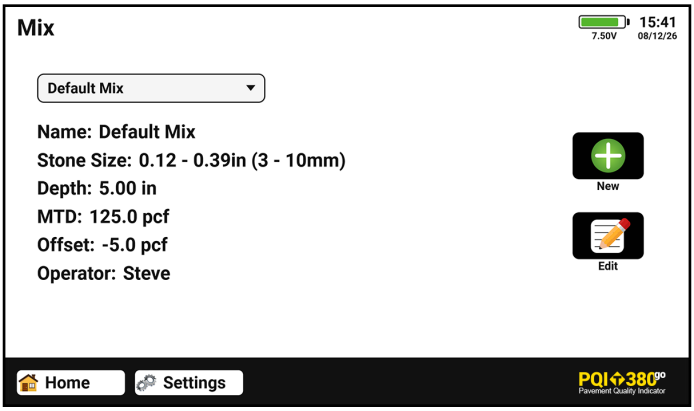


PQI 380+ Offset Setup

In the Edit Mix screen, verify the updated offset information for that mix. Once satisfied, tap **Accept** again to save your results and return to the Mix screen.



The adjusted value and new offset will be the same if the current offset was originally zero. If the current offset had been -1 and the adjusted value 5, then the new offset would be -6 for a PQI 380go that was reading too high.



The offset for each mix can be adjusted at any time. Readings taken prior to a specific adjustment will not reflect that adjustment.

Troubleshooting

The chart below provides guidance to a few suspect conditions.

Problem	Remedy
Buttons Sticking Frozen Software	Repress the frozen button firmly. Be sure not to drag your finger from one button to the next. If this does not work, reset the gauge by turning it off and then back on.
Incorrect Density Reading	**Clean off sensor plate** **Check Current Mix Details-** be sure you have selected the proper aggregate size and entered all details correctly **Check Calibration-** check your calculations for the adjusted value of the offset and that it was entered correctly **Call Factory**
Battery Problems Gauge Keeps Shutting Down Gauge Will Not Come On	**Make sure the battery pack is plugged in** **Check battery voltage-** plug gauge into charger and wait for the green LED to illuminate to indicate a full charge **Call Factory**

Replacement Parts

For a listing of replacement parts visit our website or contact your local distributor.

Warranty

Product Warranty

TransTech Systems, Inc. (the Company) warrants to the Purchaser that the product delivered hereunder will be free from defects in material or workmanship and be the kind and quality designated or specified in the contract or purchase order. This warranty shall apply only to defects appearing within one (1) year from the date of shipment by the Company.

If the product delivered hereunder does not meet the above warranty and if the Purchaser promptly notifies the Company, the Company shall thereupon correct any defect, including nonconformance with the specifications, either (at the Company's option) by repairing any defective or damaged parts of the product, replacing the product, or by making available the necessary repaired or replacement parts.

The liability of the Company under this warranty, for any loss, whether the claim is based on contract or negligence, shall not in any case exceed the cost of correcting defects in the product as herein provided, and upon the expiration of the warranty period, all such liability shall terminate. The foregoing shall constitute the exclusive remedy of the Purchaser and the exclusive liability of the Company. The foregoing warranty is exclusive and in lieu of all other warranties, whether written, oral, implied or statutory.

No warranty of merchantability or of fitness for purpose shall apply. Unauthorized service shall void this warranty.

Warranty

TransTech Systems Product Non-Warranty Return Policy (Domestic Sales Only)

Non-warranty returns for TransTech Systems Inc. products must be made within ten (10) days from the original date of shipment, unless otherwise indicated. Returned products must be in the original packaging, unused and in undamaged condition. Proof of purchase is required. Upon receipt of the product TransTech will inspect the product to the above mentioned criteria. TransTech Systems reserves the right to apply a restocking fee.

Unused products will be issued a credit to the Purchaser's account that was used to purchase the product. TransTech will not credit prepaid shipping cost. The original packing slip or invoice is required to be sent back with the product to be returned.

The Purchaser is responsible for shipping the product back to TransTech Systems, carefully package the item(s) and include the packing slip and return manufacturing authorization number on the package. Prepay shipping is required – TransTech will not accept C.O.D.s.

Returns will be credited within 10 working days.

Proper Process of Warranty or Non-Warranty Shipments to TransTech Systems

For product returns (warranty or non-warranty), please follow the instructions below to assure prompt handling:

Contact TransTech Systems by phone, email or through our website to request a Repair Ticket and detailed return shipping procedures.

Indicate the reason for returning the product.

For Warranty Returns, Purchaser is responsible for shipping to TransTech's office. TransTech will pay ground shipping return to the Purchaser.

Measurement Table

Company _____ Date _____

Job Site _____

Mix _____ MTD _____

Core Comparison Calibration Work Sheet

1. Enter the core density readings and the average PQI 380go readings for each location.

2. Subtract the PQI readings from the core density for each location.

3. Add the differences for each location = total difference.

4. Calculate the average difference (total difference ÷ 5). This is the adjusted value for the offset.

Location	1	2	3	4	5
Core Density					
PQI Readings					
Difference					

Total Difference	
Average Difference (Adjusted Value Offset)	

Measurement Table

Company _____ Date _____

Job Site _____

Mix _____ MTD _____

Core Comparison Calibration Work Sheet

1. Enter the core density readings and the average PQI 380go readings for each location.

2. Subtract the PQI readings from the core density for each location.

3. Add the differences for each location = total difference.

4. Calculate the average difference (total difference ÷ 5). This is the adjusted value for the offset.

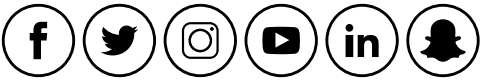
Location	1	2	3	4	5
Core Density					
PQI Readings					
Difference					

Total Difference	
Average Difference (Adjusted Value Offset)	

TransTech Systems, Inc.

900 Albany Shaker Road, Suite 2
Latham, NY 12110
Phone: 518-370-5558
Toll Free: 1-800-724-6306
Fax: 518-370-5538
Email: inquiries@transtechsys.com
Web: www.transtechsys.com

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